

# CARTER RODRIGUEZ

Computer Engineer | Hardware & Software Integration | Optimization

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## Summary

With over 8 years in computer engineering, specializing in hardware and software integration, adept at optimizing system performance. Utilizes strong problem-solving skills to collaborate effectively, evidenced by leading a 20% increase in system efficiency at a prior role. Excels in detailed documentation and fostering cross-functional team synergy.

## Skills

C++ • Python • Computer Architecture • EDA Tools • FPGA Design • Embedded Systems

## Experience

### Aegis Computational Solutions

Chicago, IL

#### Senior Computer Engineer

01/2025 - Present

- Designed and implemented complex computer systems that increased computational speed by 30% over previous models using advanced digital design techniques.
- Conducted performance analysis leading to a 20% increase in overall system efficiency by fine-tuning hardware components.
- Collaborated with cross-functional teams, fostering innovative ideas and accelerating project timelines by 15%.
- Developed detailed technical documentation that improved team comprehension and reduced project errors by 10%.
- Spearheaded troubleshooting initiatives, reducing system downtime by 25% through proactive solution implementation.

### Vertex Technology Group

Evanston, IL

#### Hardware Engineer

04/2021 - 12/2024

- Led validation and testing protocols ensuring compliance with industry standards, achieving a 98% success rate across projects.
- Collaborated with software teams to integrate hardware solutions effectively, resulting in a 15% enhancement in system integration speed.
- Implemented security measures protecting sensitive data, thus preventing potential breaches and enhancing system reliability by 20%.
- Effectively communicated complex engineering concepts to non-technical stakeholders, enhancing understanding and aligning project goals.

### TechnoCore Manufacturing

Naperville, IL

#### Embedded Systems Specialist

01/2018 - 03/2021

- Developed and optimized embedded systems leading to a 25% reduction in power consumption and increased device performance.
- Adopted new methodologies in computer architecture that resulted in a 20% enhancement in system responsiveness and reliability.
- Utilized EDA tools to streamline design processes, cutting development time by 10% while maintaining high quality standards.
- Collaborated with team members to troubleshoot software issues, improving overall system functionality by 15%.

## Education

### University of Illinois at Urbana-Champaign

Champaign, IL

#### Bachelor of Science in Computer Engineering

01/2014 - 01/2018

## Projects

### Open Source Embedded System Project

Led a team to develop an embedded system for real-time data processing, enhancing data accuracy by 15%. GitHub: [github.com/carodriguez/embedded-system](https://github.com/carodriguez/embedded-system)

### Enhanced Security Protocol Project

Created an open-source security protocol that reduced vulnerability risks in small-scale networks by 30%. GitHub: [github.com/carodriguez/security-protocol](https://github.com/carodriguez/security-protocol)

## Key Achievements



### Increased Computational Speed

Successfully increased computational speed by 30% through innovative hardware design.



### System Efficiency Improvement

Led a project that increased overall system efficiency by 20% through precise performance analysis and enhancements.



### System Downtime Reduction

Implemented troubleshooting protocols that reduced system downtime by 25%, maintaining continuous operations.

## Interests

### Computer System Optimization

Driving innovation in computing systems to advance technology and improve user experiences.